

Practical 11 - Shrinkage and variable selection - ridge and lasso regression

Statistical Models for Discrete Outcomes

London School of Hygiene & Tropical Medicine

Datasets required: `SAheart.csv`

In this practical we will explore ridge and lasso regression using the South Africa data used in the previous practical.

Question 1

Load the dataset and fit the same logistic regression model as in the previous practical for the `chd` outcome, with covariates `sbp tobacco ldl famhist obesity alcohol age`, saving the estimates into an object called `logitmle`.

We will next fit the model using ridge regression, using the R package `glmnet` (which you will need to install).

Question 2

Run the following code to fit the model using ridge regression. You will need to examine the documentation for `glmnet` to find out what value of `alpha` you need for ridge regression.

```
library(glmnet)
ridge_fit <- glmnet(model.matrix(logitmle),
                    SAheart$chd, family = "binomial",
                    alpha=)
```

Question 3

Run `plot(ridge_fit)`. What is the plot showing? Take a look at `?plot.glmnet` to see how you can change the plot.

Question 4

Use the `cv.glmnet` function to perform 10 fold cross-validation for the ridge regression. If you want to reproduce the results in the solutions, use `set.seed(789)` before you run `cv.glmnet`.

Plot the object that `cv.glmnet` returns. What does the plot show? Page 5 of the `glmnet` package [vignette](#) gives further details.

Question 5

Examine the documentation for `coef.glmnet`. Call this function with your cross-validation object from the previous question to examine the estimated coefficients at the value of λ found to be the best from cross-validation. Compare the coefficients with those from question 1.

Question 6

Examine the documentation for `predict.cv.glmnet`. Then use `predict` applied to your cross-validation object to create the fitted probability values of the CHD outcome when λ is set to the value found to be best by cross-validation. When calling `predict` you can use `type="response"` to request fitted probabilities.

Compare the fitted values from ridge regression at this value of λ to the fitted values from the MLE logistic regression in question 1.

Question 7

We will now explore lasso regression.

Fit the same logistic model using lasso regression with `glmnet`. Produce the plot of coefficients against λ . What has changed compared to the earlier ridge regression?

Question 8

Use cross-validation for lasso with 10 folds and `set.seed(789)` to find the value of λ that minimizes the deviance. Use `coef.glmnet()` to find which covariates have been retained at this value of λ , and compare with the earlier MLE and ridge regression estimated coefficients.

Question 9

Use the following code (from the last practical) to split the **SAheart** dataset into training and test subsets:

```
set.seed(592723)
n <- dim(SAheart)[1]
train_idx <- sample(seq_len(n), size = 0.75 * n)
trainData <- SAheart[train_idx, ]
testData <- SAheart[-train_idx, ]
```

Next, fit the logistic model using the *training data* with:

- MLE with `glm`
- Cross-validated ridge regression with `cv.glmnet`
- Cross-validated lasso regression with `cv.glmnet`

Then use `assess.glmnet` to assess the performance of the ridge and lasso models. Lastly, you can use the `PROC` package and code from the previous practical to assess the performance of the logistic model fitted by MLE. Compare the performance of all three.